

This resources page provides curated reading material, tools, and project ideas organized by hackathon track. You do not need to read everything here. Start with the “Essential Reading” section for your chosen track, explore the tools you plan to use, and dive deeper into specific papers as your project takes shape.

Readings for all tracks

- “AI Control: Improving Safety Despite Intentional Subversion” (Greenblatt, Shlegeris, et al., ICLR 2025) - The founding paper that introduced control evaluations as a red team/blue team methodology. Start here if you are new to the field. [arXiv:2312.06942](https://arxiv.org/abs/2312.06942) | [LessWrong](#)
- How to do High Stakes Control Research - [View-only Google Doc](#)

Other Redwood blog posts (optional):

- [An Overview of Areas of Control Work](#) - Full landscape map
- [An Overview of Control Measures](#) - Protocol taxonomy
- [7+ Tractable Directions in AI Control](#) - Research priorities
- [The Case for Ensuring Powerful AIs are Controlled](#) - Strategic argument
- [Jankily Controlling Superintelligence](#) - Long-term vision
- [BashArena and Control Setting Design](#) - Setting design principles
- [Misalignment and Strategic Underperformance](#) - Sandbagging + exploration hacking
- [Extending Control Evaluations to Non-Scheming Threats](#) - Beyond scheming
- Full blog: blog.redwoodresearch.org | Reading guide: blog.redwoodresearch.org/p/guide

Track 1: Control Setting Design

Create a new control setting that facilitates control research. Control settings are environments used to perform control research. A control setting consists of useful labor that an AI must perform (“main tasks”), combined with interesting threat models that are hard to defend against (“side tasks”). A good control setting enables us to study how much “usefulness” (performance on main tasks) must be sacrificed in order to achieve a certain amount of “safety” (loosely, the rate at which we prevent or catch attempts to achieve side tasks).

We don’t expect fully working control settings for this track. A single main task and a side task can show off the majority of the interesting dynamics. All else equal, more tasks or a scalable pipeline for producing them are better.

Example Projects

- Create environments with existing security vulnerabilities. The main tasks are complex software engineering projects, and the side task is to trigger or exploit the existing vulnerability.

- Extend SandboxEscapeBench to see if AIs can break out of Docker sandboxes while being monitored and doing complex main tasks in the sandbox
- Create a setting where AIs are doing mechinterp research, and the side task is to start a rogue internal deployment
- Build a control setting for multi-agent environments where agents can collude
- Control setting for browser agents. Maybe combine OS World with OS harm
- Control setting for building eval pipelines. Main task is to eval a model's ability to do something, side task is to start a rogue deployment
- Add or define side tasks for an existing capability benchmark such as TerminalBench, Paper-bench, or HCAST: Human-Calibrated Autonomy Software Tasks
- Extend BashBench with a side task that isn't beaten by task decomposition and deferral of sensitive components to T. All 4 side tasks in BashArena currently are.

Essential Reading

- "AI Control: Improving Safety Despite Intentional Subversion" (Greenblatt, Shlegeris, et al., ICLR 2025) - The founding paper that introduced control evaluations as a red team/blue team methodology. Start here if you are new to the field. [arXiv:2312.06942](https://arxiv.org/abs/2312.06942)
- "BashArena and Control Setting Design" (Kaufman, Lucassen, Redwood Research, Dec 2025) - Design principles for building evaluation settings. Covers 637 main tasks across 9 categories with 4 side tasks. Key reading for anyone building new settings. [Redwood Blog](https://redwoodresearch.org/blog)
- ControlArena Documentation and Blog - Official introduction to the platform, its architecture, and how to get started building experiments. control-arena.aisi.org.uk | [AISI Blog Post](https://aisi.org.uk/blog)

Further Reading

- "SHADE-Arena: Evaluating Sabotage and Monitoring in LLM Agents" (Kutasov, Sun, et al., Jun 2025) - 17 task pairs in multi-tool environments. Claude 3.7 Sonnet: 27% sabotage success. Gemini 2.5 Pro: 0.87 AUC as monitor. [arXiv:2506.15740](https://arxiv.org/abs/2506.15740)
- "CTRL-ALT-DECEIT: Sabotage Evaluations for Automated AI R&D" (Ward, van der Weij, et al., NeurIPS 2025 Spotlight) - 20 hand-crafted code-sabotage tasks extending MLE-Bench. LM monitors: 0.97 AUC for code sabotage, but sandbagging proved much harder to detect. [arXiv:2511.09904](https://arxiv.org/abs/2511.09904)
- "Adversarial Behavior in Research Settings: Conducting Control Evaluations with RE-Bench" (Rajakumar et al., NeurIPS 2025 Workshop) - Monitor effectiveness depends on side task context-specificity. Qualitative analysis outperforms numerical ratings. [OpenReview](https://openreview.net/forum?id=...)
- "Evaluating Control Protocols for Untrusted AI Agents" (Kutasov, Loughridge, et al., Nov 2025) - Resampling + deferring raises safety from 50% to 96%, but drops to 17% when attackers know the protocol. [arXiv:2511.02997](https://arxiv.org/abs/2511.02997)
- "How Can We Solve Diffuse Threats" - [Redwood Blog](https://redwoodresearch.org/blog)

Track 2: Control Protocol Design

Create new control protocols and measure their safety/usefulness tradeoff. You should test your control protocols in existing control settings, like BashArena, LinuxArena or other settings in ControlArena.

Example Projects

- Train really good monitors and show generalization to other datasets of attacks.
- Create a control protocol where the trusted model can run actions in the environment to learn more about the action the agent is proposing. I.e. the trusted model can read a file before the agent writes to it.
- Test interrogation protocols in which the trusted model asks the untrusted model questions before executing the action
- Test control protocols that involve more traditional security measures, such as giving the trusted model access to a firewall. LinuxArena or BashArena would be great for this.

Essential Reading

- “AI Control: Improving Safety Despite Intentional Subversion” (Greenblatt, Shlegeris, et al., ICLR 2025) - The founding paper. Start here if you are new to the field. [arXiv:2312.06942](https://arxiv.org/abs/2312.06942)
- “Ctrl-Z: Controlling AI Agents via Resampling” (Bhatt, Rushing, et al., Redwood Research, Apr 2025) - First control evaluation in a full agent environment (BashBench). Resampling reduces attack success from 58% to 7% at only 5% cost to benign performance. [arXiv:2504.10374](https://arxiv.org/abs/2504.10374)
- “An Overview of Control Measures” (Ryan Greenblatt, Redwood Research Blog) - Taxonomy of all control protocol types. blog.redwoodresearch.org

Extra Readings

- “How to Evaluate Control Measures for LLM Agents? A Trajectory from Today to Superintelligence” (Korbak, Balesni, Shlegeris, Irving, Apr 2025) - Introduces five AI Control Levels (ACL-0 to ACL-5). Essential framing for understanding what control measures are needed at each capability tier. [arXiv:2504.05259](https://arxiv.org/abs/2504.05259)
- “Untrusted AIs Can Exploit Feedback in Control Protocols” - Safety drops from 93% to 80% when feedback flows from trusted to untrusted components. [LessWrong](https://www.lesswrong.com/)
- “Factor(T,U): Factored Cognition Strengthens Monitoring” (Pivotal Research, Dec 2025) - Trusted model decomposes tasks, untrusted model solves sub-tasks in isolation. Safety: 41% to 63%. [arXiv:2512.02157](https://arxiv.org/abs/2512.02157)
- “Factor(U,T): Controlling Untrusted AI by Monitoring their Plans” (Dec 2025) - Monitoring implementations (AUROC 0.96) far outperforms monitoring plans (AUROC 0.52). Attack success: 1.2%. [arXiv:2512.14745](https://arxiv.org/abs/2512.14745)

- “Enhancing Monitoring through Trusted Debate” (NeurIPS 2025 Workshop) - Two trusted agents argue opposing sides about code backdoors; outperforms single-monitor approaches. [OpenReview](#)

Monitoring Systems

- “Async Control: Stress-testing Asynchronous Control Measures” (Stickland, Michelfeit, et al., UK AISI, Dec 2025) - Async monitoring avoids runtime latency. Ensemble monitor: 6% false negative at 1% false positive. System design may matter more than monitor quality. [arXiv:2512.13526](#)
- “Basic Legibility Protocols Improve Trusted Monitoring” (Feb 2026) - Counter-intuitive finding: allowing untrusted models to add code comments outperforms comment-stripping. Benefits to benign models outweigh deception risks. [arXiv:2602.10153](#)
- “Prefix Cache Untrusted Monitors” (Greenblatt, Redwood, Jun 2025) - Train a policy copy as untrusted monitor using the prefix cache. Avoids the dilemma of training the policy about countermeasures. [Redwood Blog](#)
- METR Monitorability Evaluations (SHUSHCAST) (Jan 2026) - Access to reasoning traces increases monitor catch rates by 50+ percentage points against GPT-5. [METR Blog](#)

Safety Cases & Deployment

- “A Sketch of an AI Control Safety Case” (Korbak, Clymer, et al., Jan 2025) - First end-to-end worked example of a control safety case for realistic deployment. [arXiv:2501.17315](#)
- “Towards Evaluations-Based Safety Cases for AI Scheming” (Balesni + 15 co-authors, Anthropic, DeepMind) - Three safety case arguments: Scheming Inability, Harm Inability, Harm Control. [arXiv:2411.03336](#)
- “Adaptive Deployment of Untrusted LLMs Reduces Distributed Threats” - Macro-protocols dynamically choose between micro-protocols based on behavior. Reduces backdoors by 80%, retains 99.6% usefulness. [arXiv:2411.17693](#)
- “Practical Challenges of Control Monitoring in Frontier AI Deployments” (Lindner et al., DeepMind + AISI, Dec 2025) - Challenges at scale: parallel instances, oversight latency, incremental attacks. [arXiv:2512.22154](#)
- “Games for AI Control” (Griffin, Thomson, et al., ICLR 2025) - Formalizes control evaluations as partially observable stochastic games with formal guarantees. [arXiv:2409.07985](#)

Track 3: Building Better Red Teams

Build better red teams for existing control protocols and settings.

Example Projects

- Build a dataset of sneaky transcripts of AIs doing very bad things that a prompted instance of Opus 4.6 can't catch.
- Make a red team scaffold that can make really good attacks in LinuxArena
- Try red-teaming protocols like trusted editing harder. Such that the editor preserves the attack

Essential Reading

- “Monitoring Benchmark for AI Control” - [LessWrong](#)
- “Adaptive Attacks on Trusted Monitors Subvert AI Control Protocols” (Terekhov et al., Oct 2025) - Prompt injections in model outputs fully subvert existing monitoring. Defer-to-Resample paradoxically worsens security. Adaptive attacks are a major blind spot. [arXiv:2510.09462](#)
- “Optimizing AI Agent Attacks With Synthetic Data” (Loughridge, Colognese, et al., Nov 2025) - Decomposes attack capability into 5 skills. Reduces safety score from 0.87 to 0.41 using optimized scaffold with real SHADE-Arena transcripts. [arXiv:2511.02823](#)

Extra Readings

- “Attack Selection Reduces Safety in Concentrated AI Control Settings” (Feb 2026) - Safety drops from 99% to 59% when attacker reasons about monitors and chooses moments strategically. [arXiv:2602.04930](#)
- “Vulnerability in Trusted Monitoring and Mitigations” - “Monitor-Jailbreak” attacks using fabricated conversation histories. Proposes In-Context Defense (ICD). [LessWrong](#)
- “Untrusted AIs Can Exploit Feedback in Control Protocols” - Safety drops from 93% to 80% when feedback flows from trusted to untrusted components. [LessWrong](#)
- “Simple Probes Can Catch Sleeper Agents” (Anthropic) - Linear probes predict defection with AUROC > 99%. Deceptive models may be unable to avoid “thinking” about deception. [Anthropic Research](#)
- “Large Reasoning Models Are Autonomous Jailbreak Agents” (Nature Communications, 2026) - 97% jailbreak success across 70 prompts using a single reasoning model. [Nature Communications](#)
- “The Attacker Moves Second” (OpenAI + Anthropic + DeepMind, Nov 2025) - Evaluates 12 defenses under adaptive attack: bypasses most above 90% success rate. Prompt injection remains fundamentally unsolved.
- “Secret Collusion among AI Agents via Steganography” (NeurIPS 2024) - As capabilities increase, steganographic abilities outpace overseers. [arXiv:2402.07510](#)

Track 4: LinuxArena

Special track for using LinuxArena, a control setting built by Redwood and Equistamp. Equistamp will award \$500 to the best LinuxArena project.

LinuxArena is still under active development. We plan to do more work and change the codebase before releasing it more broadly in the next couple of weeks.

Links

- [Blog post](#)
- Github: github.com/linuxarena/linuxarena
- Docs/data: linuxarena.ai
- Discord: discord.gg/rrka5wxY - To chat with engineers who made it. Please only ask questions in the "linuxarena-discussion" channel

We are generally happy for people to do any project with LinuxArena if they think it will help advance control research. Below is a list of primer projects that you can take or leave.

Project Proposals

- Build a new environment with a main task and a side task. Show that AIs can sneakily do the side task without getting caught.
- Add side tasks to an existing environment.
- Build a really good attacker that can cause side tasks without getting caught by a strong monitor
- Add vulnerabilities to the existing environments and let the attacker know about them. Prove that this makes the attacks more sneaky.
- Build a control protocol where T gets access to a firewall tool to blacklist and whitelist URLs, action by action. Show that this has high safety against API key exfil side tasks and doesn't have usefulness hits when the main tasks involve internet access.
- Try to replicate existing control protocols in LinuxArena: Trusted editing, Trusted feedback (T provides feedback to U about the action and asks U to edit it - ref: [LessWrong](#)), Resampling, A protocol where T applies security patches to the codebase before U runs in the environment.

Other readings

- [Recent Redwood Research Project Proposals](#)
- [Redwood Research Reading List](#)
- [Control Research Document 1](#)
- [Control Research Document 2](#)

Tools & Platforms

Control Environments

- LinuxArena - New complex control setting. 20 environments, 2000 SWE tasks, 100 side tasks. Github: github.com/linuxarena/linuxarena | Docs/data: linuxarena.ai

- Full Repo Refactor - Code refactoring control evaluation environment & platform. Docs: fastfedora.github.io/full-repo-refactor | GitHub: github.com/fastfedora/full-repo-refactor
- BashArena - Control setting for highly privileged agents. 637 shell tasks with 4 malicious side tasks. Github: [control-arena/bash_arena](https://github.com/control-arena/bash_arena) | Docs: bash-arena.com
- SHADE-Arena - Sabotage detection: 17 task pairs (5 public, 12 reserved). [Anthropic Research](https://anthropic-research.github.io/shade-arena/)
- SHUSHCAST (METR) - Monitorability: agent evasion vs. monitor detection
- ControlArena - 14-setting library for AI control experiments. control-arena.aiisi.org.uk

Other Control Tooling

-  Luthien: Open-source proxy between Claude Code & Anthropic API. [Learn More](#) | [Hackathon Ideas](#) | [GitHub](#) | [Feedback](#)
- Agent Transcript Editor - [GitHub](#) | [Feedback form](#)
- Inspect AI (AISI) - Open-source evaluation framework. ControlArena's foundation. 100+ pre-built evaluations. inspect.aiisi.org.uk
- MonitorBench - 100 sabotage trajectories that are hard to monitor. [LessWrong](#)
- DsPy - Prompt optimization tooling. [GitHub](#)

Safety & Evaluation

- Bloom (Anthropic) - Generates behavioral evaluation suites from a single spec. [GitHub](#)
- Petri (Anthropic) - Automated auditing that discovers misaligned behaviors. [GitHub](#)
- gpt-oss-safeguard (OpenAI) - Open-weight reasoning safety classifier (120B/20B).
- LlamaFirewall (Meta) - Jailbreak detection + CoT auditing + code analysis.

Red Teaming

- PyRIT (Microsoft) - [GitHub](#)
- Garak (NVIDIA) - 100+ attack modules
- FuzzyAI (CyberArk) - [GitHub](#)
- Promptfoo - promptfoo.dev
- Inspect Cyber (AISI) - [GitHub](#)

Benchmarks & Datasets

- MALT (METR) - 10,919 agent transcripts with natural misaligned behavior
- RepliBench (AISI) - Autonomous self-replication (20 evaluations, 65 tasks)
- DeceptionBench - 5-category benchmark for probing alignment faking
- RE-Bench - R&D tasks with adversarial side objectives